

Feature-resolved attention in the constrained-belief-update program: gauge, calculus, and every edit predicted *ex ante* in **P2**'s and **P1**'s own worlds

FRA sprint, Phase 1 — a formal note extending J. Tahir's one-layer analysis (**P3**)

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Abstract

We carry the feature-resolved attention (FRA) programme into the constrained-belief-update setting of *Piotrowski et al.* (**P2**) and *Tahir* (**P3**). Working in **P3**'s model class and **P2**'s softmax instantiation, and importing the QK gauge theorem and OV path calculus of the two prior-sprint notes ([prior-A], [prior-B]), we ask which FRA objects are computation and which are gauge, and whether every attribution and every edit can be predicted *ex ante* from the constrained-belief theory. The answer, established by exact enumeration over all $3^{10} = 59049$ length-10 Mess3 sequences, is sharp. The FRA-QK (score) side is a *gauge coordinate*: at a constrained-belief optimum the token $\times$ token score block is additively separable and consists entirely of gauge coordinates, and on the faithful **P2** model a loss-preserving weight dial (bit-identical behaviour, max logit change 7×10^{-7}) swings a fixed FRA-QK cut's cross-entropy effect by $140\times$, with the minimum sitting exactly where a measured pedestal crosses zero and an irreducible residue at the $\sim 10^{-5}$ -nat scale the loss never sees. The FRA-OV (transported content) side is the real handle: the per-(token, lag) attribution is $\zeta^{d-s}\mathbf{g}(z)$ up to a softmax-forced normalization warp $\lambda_d = 1/(1 - \zeta^d)$; severing a token's channel drives its belief-tracking linearly to zero at $c^* = 1/\rho$, with the slope ρ_{eff} predicted from clean-model statistics alone to *four significant figures* on all five measured channels, and collapses the intermediate fractal along $\mathbf{g}(z)$. The sparse dictionary that names the features is itself the binding constraint: a TopK SAE learns a minimal contrast code in which one of three token channels does not exist as a cut set. For the negative-eigenvalue case only the head-sum $\sum_h \mathbf{A}_{d,s}^{(h)} \mathbf{V}^{(h)} \mathbf{e}_{z_s} = \zeta^{d-s}\mathbf{g}(z_s)$ is loss-pinned; per-head attributions are a gauge family *not* freely realizable under softmax ($O(\varepsilon)$ generic, $O(1)$ at parity-empty rows). Every claim carries an honesty tag; we close with the bridge to **P1**'s factored world and an honesty ledger.

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1 Introduction

1.1 What feature-resolved attention is

Fix the one-layer attention circuit of **P3** (Eq. (2) below): the destination residual receives a *direct* contribution $\mathbf{W}_U \mathbf{W}_E \mathbf{x}_d$ and an *attention-mediated* contribution $\sum_{s \leq d} \mathbf{A}_{ds} \mathbf{W}_U \mathbf{W}_{OV} \mathbf{W}_E \mathbf{x}_s$. Feature-resolved attention (FRA) is the decomposition of the two attention-internal objects in a chosen feature basis of the residual stream:

- **FRA-QK** resolves the pre-softmax score $\sigma(d, s) = \mathbf{q}_d \cdot \mathbf{k}_s$ into *feature* $\times$ *feature* terms: which query feature of the destination reads which key feature of the source. In a basis $\mathbf{x}_s = \mathbf{e}(z_s) + \mathbf{p}_s$ (token feature plus positional feature) the score splits exactly into four blocks (pos $\times$ pos, pos $\times$ tok, tok $\times$ pos, tok $\times$ tok); the token $\times$ token block is the FRA-QK *feature-content* score.

- **FRA-OV** resolves the transported content $\mathbf{A}_{ds}\mathbf{W}_U\mathbf{W}_{OV}\mathbf{W}_E\mathbf{x}_s$ by *source feature and lag*: which feature, copied from how many tokens back, is written into the destination. It factorizes as (a scalar lag profile) $\times$ (a token-indexed content direction).

An FRA *attribution* reads a coefficient off one of these resolved objects; an FRA *edit* (or *cut*) removes a named feature-term at a gain c ($c = 1$ is exact severing, $c > 1$ counter-injects) and asks what the model then does. The programme’s central question is whether these attributions and edits measure the network’s *computation* or merely a *gauge* choice — a coordinate that a loss-preserving reparameterization can move at will.

1.2 The question, in P2’s and P3’s own world

P2 shows that a one-layer transformer trained on a single Mess3 process represents, at the post-attention residual, the *constrained belief update*

$$\mathbf{r}_1(d) = \boldsymbol{\pi} + \sum_{s=1}^d \zeta^{d-s} \mathbf{g}(z_s), \quad (1)$$

a lag-weighted sum of per-token displacement vectors $\mathbf{g}(z_s)$ in a two-dimensional belief plane, with a single spectral rate ζ . **P3** derives, for the frozen-attention MSE surrogate, the optimal $\mathbf{W} = \mathbf{W}_U\mathbf{W}_E$, $\mathbf{V} = \mathbf{W}_U\mathbf{W}_{OV}\mathbf{W}_E$ and the exact loss symmetry $\mathbf{W} \rightarrow \mathbf{W} + \mathbf{u}\mathbf{1}^\top$, $\mathbf{V} \rightarrow \mathbf{V} - \mathbf{u}\mathbf{1}^\top$. The prior sprint built, on a separate toy platform, a theory of FRA: a QK gauge theorem (which score edits are gauge) and an OV path calculus (which content edits obey a tracking law). This note asks the concrete question:

In P2’s exact setting, which FRA objects are computation and which are gauge, and can every attribution and every edit be predicted ex ante from the constrained-belief theory?

The headline, established by exact enumeration (no sampling error anywhere in Phase 1):

1. **FRA-QK is gauge** (§4). At a constrained-belief optimum the token $\times$ token score block is additively separable and consists entirely of gauge coordinates; its measured cut effect on the faithful **P2** model is a 140 $\times$ V-shape as a loss-preserving dial is turned, minimized where a measured pedestal crosses zero, with an irreducible residue at the noise floor the loss never sees.
2. **FRA-OV is the handle** (§5). The transported-content attributions are the constrained-belief per-source terms; severing a token’s channel removes exactly the predicted content, tracking response is linear in the gain, and the *ex-ante* slope ρ_{eff} — from clean-model statistics only — matches the measured tracking null to four significant figures on all five channels.
3. **The dictionary is the binding constraint** (§6). A TopK SAE learns a minimal contrast code in which one of three token channels does not exist as a latent set; every intervention number remains predictable, but only through the code’s mixing matrix.
4. **The two-head hinge** (§7). For $\zeta < 0$ only the head-sum attribution is loss-pinned; the per-head decomposition is a gauge family that is *not* freely realizable under softmax, obstructing clean single-gain interventions.

1.3 Relation to P1, P2, P3, and the prior notes

This is an *extension of P3*. We adopt its notation throughout (bold vectors $\mathbf{x}, \boldsymbol{\pi}, \mathbf{u}$; matrices $\mathbf{T}, \mathbf{A}$; token-labeled $\mathbf{T}^{(x)}$; the effective $\mathbf{W}, \mathbf{V}$; the $f(\mathbf{X}) = \mathbf{W}\mathbf{X} + \mathbf{V}\mathbf{X}\mathbf{A}^\top$ decomposition and its loss symmetry). We instantiate **P2**’s softmax App.-B model faithfully and use its constrained-belief geometry as ground truth. We import **P1**’s attention-head-unit results — the $k=0$ diagonal/skip redundancy (**P1** Eq. 28), the “only the head-aggregate is constrained” principle, effective subspace attention, the conic head-count, and the rich/lazy init dependence (**P1** App. G) — as stated results, verified to hold in the instantiation but not re-proved. From the two prior-sprint notes we import: the QK gauge characterization, gauge coordinates, gauge families G1–G3, cut

calculus, and the separability boundary ([prior-A]); and the normalization-warp, sink realization, and two-head warp results ([prior-B]). Imported statements are quoted compactly with attribution and their premises checked in the **P2** instantiation; the new content is the instantiation, the closed forms specialized to Eq. (1), and the exact measured verification. The audience is the authors of **P1/P2/P3**.

1.4 Honesty tags

Every quantitative or structural claim carries one of four tags, used exactly as in the sprint’s derivation ledger:

- [exact] — an algebraic identity, no approximation.
- [exact given ansatz] — exact once **P2**’s spectral ansatz (pattern $\propto \zeta^{d-s}$, values and embeddings $\propto \mathbf{g}(z)$) is *posited*; the ansatz itself is **P2**’s empirical fit, not derived from the CE loss (this is **P3**/[prior-A]’s Conjecture Z1).
- [approximate, error $O(\cdot)$] — a controlled approximation with the stated leading error.
- [conjecture] — plausible, not proven here.

Measured numbers are traceable to a named JSON artifact under `sprint/code/phase1/out/` (cited in footnotes); all are exact enumeration unless a sampled estimator is named.

2 Setting

2.1 The model class (**P3**)

We study **P3**’s one-head attention-only predictor

$$f(\mathbf{X})_{t,:} = \mathbf{W}_U \mathbf{W}_E \mathbf{x}_t + \sum_{s \leq t} \mathbf{W}_U \mathbf{A}_{ts} \mathbf{W}_{OV} \mathbf{W}_E \mathbf{x}_s, \quad f(\mathbf{X}) = \mathbf{W} \mathbf{X} + \mathbf{V} \mathbf{X} \mathbf{A}^\top, \quad (2)$$

with effective matrices $\mathbf{W} = \mathbf{W}_U \mathbf{W}_E$ (direct path) and $\mathbf{V} = \mathbf{W}_U \mathbf{W}_{OV} \mathbf{W}_E$ (context path) and $\mathbf{A}$ a causal row-stochastic matrix. **P3** analyses the frozen-attention optimum under the MSE surrogate and establishes the $|\mathcal{V}|$ -dimensional loss symmetry

$$\mathbf{W} \rightarrow \mathbf{W} + \mathbf{u} \mathbf{1}_{|\mathcal{V}|}^\top, \quad \mathbf{V} \rightarrow \mathbf{V} - \mathbf{u} \mathbf{1}_{|\mathcal{V}|}^\top \quad (\mathbf{u} \in \mathbb{R}^{|\mathcal{V}|} \text{ arbitrary}), \quad (3)$$

the ambient degeneracy that, resolved in a feature basis, becomes the FRA-QK gauge of §4 and the $k=0$ skip/diagonal split of §3. **P2** trains this class with softmax attention and CE loss; we keep **P3**’s function-space vocabulary and add the softmax scores only where FRA-QK needs them.

2.2 The P2 softmax instantiation (App. B, faithful)

We reproduce **P2**’s App.-B configuration exactly (`common.py`): a `TransformerLens HookedTransformer`, `n_layers=1`, $d_{\text{model}} = 64$, $d_{\text{mlp}} = 256$, GELU, LayerNorm, causal attention, $d_{\text{vocab}} = 3$, *no* BOS, learned additive positional embeddings, context length $L = N_{\text{ctx}} = 10$, cross-entropy loss, Adam (10^{-4} , no weight decay), $\approx 15\text{M}$ tokens. Two faithful configurations:

- **Config A**: $x = 0.15$, $\alpha = 0.6$, $\zeta = 1 - 3x = +0.55$, one head ($d_{\text{head}} = 64$).
- **Config B**: $x = 0.50$, $\alpha = 0.6$, $\zeta = 1 - 3x = -0.50$, two heads ($d_{\text{head}} = 32$), anti-parallel OVs.

The residual at source s is $\mathbf{x}_s = \mathbf{e}(z_s) + \mathbf{p}_s$; the pattern is $\mathbf{A}_{d,s} = \text{softmax}(\sigma(d, \cdot))_s$ with bilinear scores $\sigma(d, s) = (\mathbf{W}_Q \phi(\mathbf{x}_d) + \mathbf{b}_Q) \cdot (\mathbf{W}_K \phi(\mathbf{x}_s) + \mathbf{b}_K) / \sqrt{d_h}$, value $\mathbf{v}_s = \mathbf{W}_{OV} \mathbf{x}_s$, and frozen-LN map ϕ . The belief geometry

lives at the post-attention, pre-MLP residual ($\text{resid}_{\text{mid}}$, $\mathbf{P2}$'s $x_d^{(\text{mid})}$), $x_d^{\text{mid}} = x_d + c_d$, $c_d = \sum_{s \leq d} A_{d,s} v_s$, read out by a *linear* map $\mathbf{R} : \mathbb{R}^d \rightarrow \mathcal{P}$ to the belief plane. The MLP performs the subsequent nonlinear warp from the constrained fractal to the full Bayesian geometry ($\mathbf{P2}$, “Post-MLP”); it is outside the FRA-of-attention analysis and not modeled here. LayerNorm is treated as a frozen, content-independent per-position linear scale Φ (imported Assumption (A2)); the residual token-dependent LN leak is absorbed into the measurable *content leak* ε (imported Definition, [prior-A]).

2.3 Exact-enumeration methodology

Remark 1 (the analysis distribution is exact). Every Phase-1 number is *exact*. The analysis set is all $|\mathcal{V}|^L = 3^{10} = 59049$ length- L input sequences weighted by their exact process probability $P(z_{1:L})$, so all expectations (patterns, belief clouds, CE, tracking covariances, cut effects) are computed with no sampling error. Training uses freshly sampled data; all geometry and FRA analysis uses the enumeration. This mirrors $\mathbf{P2}$'s own exact treatment and lets us state cut effects down to the 10^{-6} -nat scale without error bars. A numerical residual quoted as “ $k \cdot 10^{-m}$ ” is float precision or a genuine model deviation, never estimation error.

2.4 The Mess3 process algebra

Token-labeled transition operators (standard convention, `mess3.py`) are $T_{ij}^{(x)} = t(i, j) e(x | j)$ with $t = (1 - 2x)\mathbf{I} + x(\mathbf{1}\mathbf{1}^\top - \mathbf{I})$ and $e(x | j) = \alpha$ if $x = j$ else $(1 - \alpha)/2$. All of the following are [exact] (checked in `mess3.py`): $\mathbf{T} := \sum_x \mathbf{T}^{(x)} = t$ is row-stochastic with stationary $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and eigenvalues $\{1, \zeta, \zeta\}$, $\zeta = 1 - 3x$ doubly degenerate; $P(x) = \pi \mathbf{T}^{(x)} \mathbf{1} = \frac{1}{3}$; the conditional transition $\mathbf{T}^{|x} := \mathbf{T}^{(x)} / P(x)$. The load-bearing object is the *token displacement*

$$\mathbf{g}(x) := \pi \mathbf{T}^{|x} - \pi \in \mathbb{R}^3, \quad \mathbf{g}(x) \cdot \mathbf{1} = 0, \quad \sum_x \mathbf{g}(x) = 0. \quad (4)$$

The spectral projectors are $\mathbf{T} = \mathbf{T}_1 + \zeta \mathbf{P}_\zeta$ with $\mathbf{T}_1 = \mathbf{1}\pi$ and $\mathbf{P}_\zeta = \mathbf{I} - \mathbf{1}\pi$ ($\mathbf{P}_\zeta \mathbf{1} = 0$, $\pi \mathbf{P}_\zeta = 0$, $\mathbf{P}_\zeta^2 = \mathbf{P}_\zeta$). The *belief plane* $\mathcal{P} := \{v \in \mathbb{R}^3 : v \cdot \mathbf{1} = 0\}$ has $\dim \mathcal{P} = 2$; all three $\mathbf{g}(x)$ lie in $\mathcal{P}$ and (since $\sum_x \mathbf{g}(x) = 0$) span it, so any two are a basis, and $\mathcal{P} = \text{im}(\mathbf{P}_\zeta)$.

Lemma 1 (spectral identity). [exact] For every token x and integer $n \geq 0$, $\pi \mathbf{T}^{|x} \mathbf{T}^n - \pi = \zeta^n \mathbf{g}(x)$.

Proof. $\pi \mathbf{T}^{|x} = \pi + \mathbf{g}(x)$ by (4) and $\pi \mathbf{T}^n = \pi$, so the left side is $\mathbf{g}(x) \mathbf{T}^n$. Now $\mathbf{g}(x) \cdot (\mathbf{1}\pi) = (\mathbf{g}(x) \cdot \mathbf{1})\pi = 0$, so $\mathbf{g}(x) \mathbf{T} = \zeta \mathbf{g}(x) \mathbf{P}_\zeta = \zeta \mathbf{g}(x)$, whence $\mathbf{g}(x) \mathbf{T}^n = \zeta^n \mathbf{g}(x)$. $\square$

Summing the per-source terms gives the constrained belief update in spectral form ($\mathbf{P2}$'s spectral constrained-belief equation, instantiated), i.e. Eq. (1):

$$\mathbf{r}_1(z_{1:d}) = \pi + \mathbf{u}_d, \quad \mathbf{u}_d := \sum_{s=1}^d (\pi \mathbf{T}^{|z_s} \mathbf{T}^{d-s} - \pi) \stackrel{\text{Lem. 1}}{=} \sum_{s=1}^d \zeta^{d-s} \mathbf{g}(z_s) \in \mathcal{P}. \quad (5)$$

All lag-dependence sits in the scalar ζ^{d-s} ; all token-dependence in the plane vector $\mathbf{g}(z_s)$. This factorization is what makes FRA tractable and, as §4 shows, what puts Mess3 on the gauge side of the separability boundary. Numerically, $\max_x \max_{n \leq 5} \|(\pi \mathbf{T}^{|x} \mathbf{T}^n - \pi) - \zeta^n \mathbf{g}(x)\|_\infty = 1.1 \times 10^{-16}$.¹

2.5 The bridge lemma: $\mathbf{P3}$'s token statistics carry $\mathbf{P2}$'s belief vectors

The formal link between $\mathbf{P3}$'s MSE-optimal effective parameters and $\mathbf{P2}$'s belief-plane displacements is a column identity for $\mathbf{P3}$'s spectral skip-bigrams $\mathbf{M}_\lambda$ (with $(\mathbf{M}_\lambda)_{ij} = \pi \mathbf{T}^{(j)} \mathbf{P}_\lambda \mathbf{T}^{(i)} \mathbf{1}$), imported verbatim from [prior-A].

Lemma 2 (bridge; imported [prior-A]). [exact] For every nonstationary eigenvalue $\lambda \neq 1$, $\mathbf{M}_\lambda e_j = p_j \mathbf{W}_U (\mathbf{g}(j) \mathbf{P}_\lambda)^\top$, where $\mathbf{W}_U$ is $\mathbf{P3}$'s unembedding (rows $\mathbf{1}^\top (\mathbf{T}^{(i)})^\top$) and p_j the stationary probability of token j .

¹`mess3.py` self-check; $x = 0.15, \alpha = 0.6, \zeta = 0.55$.

Proof (imported). $(\mathbf{M}_\lambda)_{ij} = \boldsymbol{\pi} \mathbf{T}^{(j)} \mathbf{P}_\lambda \mathbf{T}^{(i)} \mathbf{1}$; write $\boldsymbol{\pi} \mathbf{T}^{(j)} = p_j(\boldsymbol{\pi} + \mathbf{g}(j))$ and use $\boldsymbol{\pi} \mathbf{P}_\lambda = 0$ for $\lambda \neq 1$ to get $(\mathbf{M}_\lambda)_{ij} = p_j \mathbf{g}(j) \mathbf{P}_\lambda \mathbf{T}^{(i)} \mathbf{1} = p_j (\mathbf{W}_U(\mathbf{g}(j) \mathbf{P}_\lambda)^\top)_i$. $\square$

Column j of $\mathbf{M}_\lambda$ is p_j times the unembedded λ -component of token j 's belief displacement: $\mathbf{P2}$'s token-space regression statistics are, in each nonstationary block, exactly $\mathbf{P2}$'s $\mathbf{g}(j)$ -vectors. This is the bridge that gives the FRA-OV attribution its $\mathbf{g}(z)$ -shape (§5).

3 The P2 ideal configuration

3.1 Ideal feature basis and pattern

Imported Assumption (A1) posits the exact residual split $\mathbf{x}_s = \mathbf{e}(z_s) + \mathbf{p}_s$; $\mathbf{P2}$'s empirical "embeddings $\parallel$ OV vectors $\propto \mathbf{g}(z)$ " is posited as the *ideal embedding* $\mathbf{e}(z) = C_e \hat{\mathbf{g}}(z)$ (a fixed linear lift $\mathcal{P} \hookrightarrow \mathbb{R}^{d_{\text{model}}}$), so the linear readout obeys $\mathbf{R} \mathbf{e}(z) = C_e \mathbf{g}(z)$ on the query/direct side and $\mathbf{R} \mathbf{W}_{OV} \mathbf{e}(z) = C_v \mathbf{g}(z)$ on the value side ($\mathbf{P2}$'s fvn: all OV vectors of a token are parallel to $\mathbf{g}(z)$). [exact given ansatz]

Definition 3.1 ($\mathbf{P2}$ ideal pattern). The pure-geometric, row-normalized lag profile is

$$\mathbf{A}_{d,s} = \frac{\zeta^{d-s}}{N_d}, \quad N_d = \sum_{s'=1}^d \zeta^{d-s'} = \frac{1-\zeta^d}{1-\zeta} \quad (1 \leq s \leq d). \quad (6)$$

This is *lag-only with d-dependent normalization* in the imported pattern-class sense; it is the normalizer entering the belief-cut formula (12).

Proposition 1 (realizability of the geometric profile; imported [prior-A]). [exact for $\zeta \in (0, 1)$] The pattern (6) is produced by a single softmax head with key-side-only, position-linear scores $\sigma(d, s) = s(-\ln \zeta) + \text{const}(d)$, i.e. $k_P(s) = s|\ln \zeta| \hat{u} + w$ (recency slope on the key side). For $\zeta < 0$, $\ln \zeta$ is undefined and no single softmax head realizes it (patterns are nonnegative); two heads split even/odd lags (§3.4). Verified: softmax of $\sigma(s) = s(-\ln \zeta)$ reproduces (6) to 1.7×10^{-16} .

3.2 The lag-0 split: diagonal attention vs skip (the $k=0$ gauge)

Grouping the $s = d$ (lag-0) term of $\mathbf{u}_d$ in (5), $\mathbf{P2}$'s diagonal-attention identity ($\mathbf{P2}$ Eq. **DiagonalAttention**), instantiated, is

$$\underbrace{\mathbf{R} \mathbf{x}_d^{(\text{pre})}}_{\text{skip}} + \underbrace{\mathbf{A}_{d,d} \mathbf{R} \mathbf{v}_d}_{\text{diagonal attention}} = \mathbf{g}(z_d), \quad (7)$$

i.e. the current-token displacement $\mathbf{g}(z_d)$ is *split* between the residual skip connection and the diagonal attention, and *only the sum is pinned by the loss*. This is precisely $\mathbf{P1}$'s $k=0$ degree of freedom ($\mathbf{P1}$ Eq. 28): $\mathbf{x}_d$ is itself a regressor, so diagonal-attention mass is collinear with the direct path. Write the *diagonal share* $a_0(d) := [\mathbf{A}_{d,d} \mathbf{R} \mathbf{v}_d / \mathbf{g}(z_d)]_{\text{along } \mathbf{g}(z_d)} \in [0, 1]$; $a_0 = 1$ puts all of $\mathbf{g}(z_d)$ on diagonal attention (forced by one-hot-frozen embeddings, $\mathbf{P1}$), $a_0 = 0$ puts all of it on the skip. $a_0(d)$ is *pure gauge*: unidentifiable from the ideal theory, measurable ex post as $\mathbf{A}_{d,d}$ times the OV image of the current token. It is the reason the attention-mediated path-share $\rho < 1$ in §5 (an OV cut removes the diagonal, not the skip). For (6), $\mathbf{A}_{d,d} = 1/N_d = (1-\zeta)/(1-\zeta^d)$, shrinking from 1 at $d = 1$ toward $1-\zeta$.

3.3 Ideal-ansatz vs softmax-realizable: the normalization warp

This is the single most important honesty point, and it is where [prior-B]'s "Finding 5" corrects a naive reading of $\mathbf{P2}$.

Proposition 2 (normalization warp; imported [prior-B], re-derived). [exact given the ansatz; realizability correction exact] With additive positional embeddings the value $\mathbf{v}_s = \mathbf{W}_{OV}(\mathbf{e}(z_s) + \mathbf{p}_s)$ has a *position-independent* token part,

so $\mathbf{R}\mathbf{v}_s = C_v \mathbf{g}(z_s) + \mathbf{h}_s$ with $\mathbf{h}_s$ token-independent. There is then *no* score assignment realizing $\mathbf{A}_{d,s} \mathbf{R}\mathbf{v}_s = \zeta^{d-s} \mathbf{g}(z_s)$ for all d . Concretely, with the pattern (6) and position-independent values the head output is a token-independent multiplicative warp of the target,

$$\mathbf{R}\mathbf{c}_d = \sum_{s \leq d} \mathbf{A}_{d,s} C_v \mathbf{g}(z_s) = \frac{C_v}{N_d} \mathbf{u}_d = \lambda_d \mathbf{u}_d, \quad \boxed{\lambda_d = \frac{C_v(1-\zeta)}{1-\zeta^d}}. \quad (8)$$

The unique C_v making $\lambda_d \rightarrow 1$ as $d \rightarrow \infty$ is $C_v = 1/(1-\zeta)$, giving $\lambda_d = 1/(1-\zeta^d) \geq 1$: a geometric overshoot $\zeta^d/(1-\zeta^d)$, largest at small d .

This is the exact quantitative content of **P2**’s *unexplained* “scalar discrepancy in the first two embedding vectors”: the exact-update reading requires per-position value magnitudes an additive positional embedding cannot supply. Verified by direct construction to 2.2×10^{-16} : for $C_v = 1/(1-\zeta)$ the first five destinations give $\lambda_d = (2.222, 1.434, 1.200, 1.101, 1.053) \rightarrow 1/(1-\zeta^d)$.

Proposition 3 (the exact-update alternative is not **P2**-realizable; imported [prior-B]). [exact given ansatz] One can hit $\mathbf{u}_d$ exactly with a *sink pattern* $\mathbf{A}_{d,1} = \zeta^{d-1}$, $\mathbf{A}_{d,s} = (1-\zeta)\zeta^{d-s}$ for $s \geq 2$ (position 1 is the BOS-less attention sink absorbing the excess mass ζ^{d-1}), together with position-dependent value magnitudes $\mathbf{R}\mathbf{v}_1 = \mathbf{g}(z_1)$, $\mathbf{R}\mathbf{v}_s = \mathbf{g}(z_s)/(1-\zeta)$. But a token-independent additive $\mathbf{h}_s$ cannot produce a token-dependent multiplicative $1/(1-\zeta)$, so this realization is *outside* the **P2** architecture. It is a “what it would take” reference, and it reproduces **P2**’s anomalously-large $\mathbf{A}_{2,1}$ / anomalously-small $\mathbf{v}_1$ observations exactly (head output = $\mathbf{u}_d$ to 1.1×10^{-16}).

3.4 The negative- ζ two-head realization (Config B)

For $\zeta < 0$ the target per-source coefficient $\zeta^{d-s} = (-1)^{d-s} |\zeta|^{d-s}$ oscillates in sign, but softmax patterns are nonnegative. **P2**’s resolution (**P2** Eq. `Ads_2head_relation`): *two heads*, head 1 carrying even lags, head 2 odd lags, with *anti-parallel OVs* $\widehat{\mathbf{R}\mathbf{v}_s^{(1)}} = -\widehat{\mathbf{R}\mathbf{v}_s^{(2)}}$, both $\propto \mathbf{g}(z_s)$, and the head-sum reproducing the oscillating coefficient,

$$\sum_{h=1,2} \mathbf{A}_{d,s}^{(h)} \mathbf{R}\mathbf{v}_s^{(h)} = (-1)^{d-s} |\zeta|^{d-s} \mathbf{g}(z_s) = \zeta^{d-s} \mathbf{g}(z_s). \quad (9)$$

Each head has its own non-negative pattern supported on its parity class, normalized *separately* by softmax. With $\kappa := \zeta^2$, the per-head normalizers are

$$N_d^{(1)} = \frac{1 - \kappa^{\lceil d/2 \rceil}}{1 - \kappa}, \quad N_d^{(2)} = |\zeta| \frac{1 - \kappa^{\lfloor d/2 \rfloor}}{1 - \kappa}, \quad N_d^{(1)} + N_d^{(2)} = \sum_{s \leq d} |\zeta|^{d-s}. \quad (10)$$

Because each head renormalizes its own rows, a key-side cut acts on the two heads through *different* normalizers — the source of the two-head hinge (§7). The warp of §3.3 applies per head with N_d replaced by $N_d^{(h)}$; the two heads overshoot by *different* factors ($\lambda_2 \approx 2\lambda_1$), so no single value scale un-warps both. [exact given ansatz]

4 FRA-QK in P2’s world

4.1 The four score blocks and the gauge coordinates

Define the score-side vectors ($1/\sqrt{d_h}$ absorbed, biases folded into the positional parts) $q_E(z) = \mathbf{W}_Q \Phi \mathbf{e}(z)$, $q_P(d) = \mathbf{W}_Q \Phi \mathbf{p}_d + b_Q$, $k_E(z) = \mathbf{W}_K \Phi \mathbf{e}(z)$, $k_P(s) = \mathbf{W}_K \Phi \mathbf{p}_s + b_K$, so $\sigma(d, s) = \mathbf{q}_d \cdot \mathbf{k}_s$ splits *exactly* into four blocks and the token $\times$ token block $Q(z, z') := q_E(z) \cdot k_E(z')$ is the FRA-QK feature $\times$ feature score mass. Bars denote token means, hats centered parts.

Assumption 1 (premises transfer to **P2**). [A0 exact for Mess3; A1 idealized; A2 error $O(\varepsilon)$] (A0) full support — every (token, position) and every ordered token pair at any two positions occurs (Mess3: every finite string has positive

probability); (A1) exact split $\mathbf{x}_s = \mathbf{e}(z_s) + \mathbf{p}_s$; (A2) content-independent frozen-LN scale, the residual token-dependent LN leak absorbed into the content leak $\varepsilon := \max_{d,z,z'} |\mathbf{q}_d \cdot (\mathbf{k}_E(z) - \mathbf{k}_E(z'))| + \max_{z,s,s'} |\hat{q}_E(z) \cdot (\mathbf{k}_P(s) - \mathbf{k}_P(s'))|$. Crucially, the ideal pattern (6) is token-independent (depends only on (d, s)), so Conjecture Z1 (lag-only optimality) is *assumed satisfied by construction* here; everything below is [exact given a token-independent-pattern configuration], the imported theorem’s exactness class.

Theorem 4.1 (characterization; imported [prior-A], instantiated). [exact given Assumption 1] The pattern is token-independent on the support for all $d \geq 2$ iff **(C1)** $\mathbf{q}_d \cdot \hat{\mathbf{k}}_E(z) = 0$ for every realized query and every z (including the diagonal token) and **(C2)** $\hat{q}_E(z) \cdot \hat{\mathbf{k}}_P(s) = 0$ for every z and realized s . Differencing (C1) over z_d gives **(C1a)** $\hat{q}_E(z) \cdot \hat{\mathbf{k}}_E(z') = 0$.

Proposition 4 (the feature $\times$ feature mass is pure gauge; imported [prior-A]). [exact given a token-independent pattern] Under (C1)–(C2),

$$Q(z_d, z_s) = \underbrace{\bar{q}_E \cdot \bar{\mathbf{k}}_E}_{\text{const}} + \underbrace{\hat{q}_E(z_d) \cdot \bar{\mathbf{k}}_E}_{\text{query-token gauge}} + \underbrace{\bar{q}_E \cdot \hat{\mathbf{k}}_E(z_s)}_{\text{key-token gauge } b(z_s)} + \underbrace{\hat{q}_E(z_d) \cdot \hat{\mathbf{k}}_E(z_s)}_{=0 \text{ by (C1a)}}. \quad (11)$$

Every token-dependent term is additively separable and is a coordinate on the gauge orbit spanned by the *pedestal* $\beta(d) := \mathbf{q}_d \cdot \bar{\mathbf{k}}_E$ (coefficient of every key-side cut), the *key read* $b(z) := \bar{q}_E \cdot \hat{\mathbf{k}}_E(z)$, the *profile share* θ (fraction of the recency slope carried on the token-query side, $\bar{q}_E \cdot \hat{\mathbf{k}}_P(s) = \theta s(-\ln \zeta) \bmod \text{row constant}$), and the *row constants* $\rho(z) := \hat{q}_E(z) \cdot (\bar{\mathbf{k}}_E + \bar{\mathbf{k}}_P)$ (inert: softmax kills row constants). The centered interaction block is gauge-invariant and, at a token-independent-pattern configuration, identically zero (C1a); hence the loss level set contains configurations where $Q \equiv 0$ and where Q dominates the score budget, with identical behaviour. **FRA-QK attribution here measures gauge, not computation.**

The gauge families (imported [prior-A], Prop. gauge) are explicit weight reparameterizations that move these coordinates through zero to arbitrary magnitude at fixed pattern, forward pass, and loss: G1 embedding re-split $\mathbf{e}(z) \rightarrow \mathbf{e}(z) + u$, $\mathbf{p}_s \rightarrow \mathbf{p}_s - u$ (exact *everywhere*, on and off distribution — the primary dial); G2 per-row-constant query updates; G2’ pedestal (key-side); G3 null-direction key updates (exact on distribution). What no dial can move is the centered interaction $\hat{q}_E(z) \cdot \hat{\mathbf{k}}_E(z')$; at the **P2** optimum it is zero, so *everything* FRA-QK measures is dialable. This is the imported boundary theorem, branch 1: the **P2** target kernel $\zeta^{d-s} \mathbf{g}(z_s) \mathbf{P}_\zeta$ is separable (lag factor token-free, content factor lag-free), so FRA-QK has zero causal handle at every minimizer — Mess3 is on the *null* side of the induction/matching boundary ([prior-A]’s parallelogram lemma).

4.2 Cut calculus at the P2 ideal (Config A)

Theorem 4.2 (cut calculus; imported [prior-A], instantiated). [exact given (C1)–(C2), $\varepsilon = 0$; else additive $O(c\varepsilon)$] With **P2** values $\mathbf{R}\mathbf{v}_s = C_v \mathbf{g}(z_s)$ and pattern (6):

1. *Key-side single-token cut of z^* at gain c .* The subtracted score is $\beta(d) \mathbb{I}[z_s = z^*]$ (pedestal universality: the same $\beta(d)$ for every token, by (C1)), with belief-plane response the exact two-valued form

$$\Delta \mathbf{R}c_d = C_v \frac{(e^{-c\beta(d)} - 1) w_{z^*}}{1 + (e^{-c\beta(d)} - 1) w_{z^*}} \left(\mathbf{g}(z^*) - \frac{\mathbf{r}_1(d) - \boldsymbol{\pi}}{N_d} \right), \quad w_{z^*} = \sum_{s \leq d, z_s = z^*} \mathbf{A}_{d,s} : \quad (12)$$

a distortion along $\mathbf{g}(z^*) - \bar{\mathbf{u}}_d$ (with $\bar{\mathbf{u}}_d = (\mathbf{r}_1(d) - \boldsymbol{\pi})/N_d$ the *normalized* constrained update), scaled by the gauge coordinate $\beta(d)$, saturating at “delete all z^* -keys and renormalize”, and *identically zero at every gain at the canonical gauge* $\beta = 0$. It corrupts the belief-state (grammar) direction and leaves aggregate content protected. Verified to 8.9×10^{-16} .

2. *Key-side full token set.* Every key carries exactly one active token feature, so the summed edit is $\delta(d, s) = \beta(d)$ for all s : a per-row constant, **exactly inert at every gain**. Single features are cuttable only because the cut breaks this sum. With leak, $\text{osc}(\delta) \leq \varepsilon$, effect $O(c\varepsilon)$. Verified 2.2×10^{-16} .
3. *Query-side cut of z^* .* Only rows with $z_d = z^*$ are affected, $\delta(d, s) = \rho(z^*) + b(z_s) + \chi(s)$; the constant is inert, the key-token piece $b(z_s)$ is the same function for every cut token (by (C1a)), and when the gauge

routes $\chi(s) = \theta s(-\ln \zeta)$ the affected rows' decay rate morphs

$$\zeta \longrightarrow \zeta^{1-c\theta} \quad \text{for rows with } z_d = z^*, \quad (13)$$

exactly and at all orders (uniform at $c\theta = 1$, reversed beyond); inert at the canonical gauge $\theta = 0$, $b \equiv 0$. Verified 1.7×10^{-16} .

4.3 Measured verification on the trained Config-A model

We now confirm this gauge classification transfers to the trained checkpoint (`out/A/model.pt`), via exact 3^{10} enumeration.²

G1 is exact to float precision. The embedding re-split $e(z) \rightarrow e(z) + u$, $p_s \rightarrow p_s - u$ with $u = -\bar{e}$ leaves every residual bit-identical: residual max-diff 3.0×10^{-8} , forward-pass logit max-diff 6.0×10^{-7} (float32 roundoff) across the whole dial $t \in [-1, 2]$. The primary gauge dial G1 is an everywhere-invariance of the trained model. [exact, verified to float32 precision]

Content leak is substantial — the trained pattern genuinely content-gates. Reproduced from the checkpoint: $\varepsilon_{C1}^{\max} = 0.7036$, $\varepsilon_{C2}^{\max} = 0.3394$, positional slope scale 0.463, so $\varepsilon_{C1}/\text{slope} = 1.52$. The token-independent-pattern premise holds only in expectation (pattern token-dependence relstd 5.9% mean, 14% max), so the $\varepsilon = 0$ cut formulas hold to $O(c\varepsilon)$ on the trained model, as the leak term predicts. [measured]

The key-side cut effect is controlled by the gauge pedestal — a 140× swing. Dialing G1 moves the mean pedestal $\bar{\beta}(t)$ linearly through zero, and the exact-enumeration ΔCE of the key-side z_0 cut at $c = 1$ is minimized *exactly where* $\beta \approx 0$:

dial t	−1.0	−0.5	0.0	0.5	1.0	1.5	2.0
$\bar{\beta}(t)$	−0.685	−0.496	−0.308	−0.119	+0.069	+0.258	+0.446
ΔCE (z_0 key cut)	1.1e-3	6.3e-4	2.8e-4	8.0e-5	8.1e-6	5.2e-5	1.9e-4

The ΔCE minimum (8.1×10^{-6} , essentially null) sits at $t = 1$ where the pedestal crosses zero and grows as $|\beta|$ grows either way — exactly (12)'s statement that the response is scaled by $\beta(d)$ and vanishes at the canonical gauge. *The same trained model, at a loss-identical G1 gauge, shows a 140× range of the same named cut's effect*; the residue at the V-bottom ($\sim 8 \times 10^{-6}$ nats) is the loss-flat scale below. [exact enumeration]

The full-token-set key cut is not more inert than a single-token cut. At $t = 0$, exact-enumeration ΔCE : z_0 cut 2.84×10^{-4} , z_1 0.98×10^{-4} , z_2 3.28×10^{-4} , *full-token-set* 2.96×10^{-4} . Theory (item 2 of Thm. 4.2) says the full set is exactly inert at $\varepsilon = 0$; here the $\varepsilon = 0.70$ leak makes the summed edit non-constant at $O(\varepsilon)$, so it is comparable to the single-token cuts. This is the **P2**-world confirmation of [prior-A]'s full-cut failure on trained models (“the premise holds only in expectation”). Honest negative result. [measured]

The θ -share is large. Measuring θ directly on the checkpoint as the token-query slope over the full profile slope gives $\theta = 0.76, 0.73, 0.68, 0.56$ for $d = 5, 6, 7, 8$: the model routes ~ 60 – 76% of the recency profile through the mean token embedding's query, far from the canonical $\theta = 0$. Via (13) this predicts a query-side cut of a token at $c = 1$ morphs $\zeta \rightarrow \zeta^{1-\theta} \approx \zeta^{0.3}$ on that token's rows — a large decay-rate distortion alongside near-null key-side full cuts, matching [prior-B]'s query/key asymmetry. θ is a gauge coordinate (G1 moves $\bar{q}_E$), not an invariant. [new measurement; θ is gauge]

²`out/A/gauge.qk.json`, cross-referenced by `scratchpad/verify-qk-transfer.py` (reads the checkpoint, writes nothing). Trained CE ladder: Bayes $1.08920 < \text{model } 1.08946 < \mathbf{r_1\text{-}ansatz } 1.08956$ — the model beats the ansatz readout.

The dial is seed-robust. The same G1 dial on seed 43 (`out/A2/gauge_wide.json`) shows the gauge story is not a seed-42 artifact, with one caveat on scale: seed 43’s dial direction has a $\sim 14\times$ weaker lever arm on the pedestal, so a wider sweep $t \in [-30, +30]$ is needed (G1 stays exact at every magnitude, identity $\leq 8.3 \times 10^{-7}$). Over it the pedestal $\beta(z_0)$ moves linearly through zero across $[-0.65, +0.73]$ and the same named z_0 key cut’s ΔCE traces a $34\times$ V-shape ($3.7 \times 10^{-3} \rightarrow 1.65 \times 10^{-4} \rightarrow 5.6 \times 10^{-3}$) whose minimum sits *exactly* at the pedestal zero-crossing — at the dial value $t = -1.7$ predicted *ex ante* from the pedestal’s linear dial response. Only the V floor is seed-dependent (1.65×10^{-4} here vs seed 42’s 8×10^{-6} : the $O(c\varepsilon)$ leak coefficient differs at matched ε); the gauge structure — linear pedestal, zero-crossing minimum, ex-ante prediction — is identical. [measured, second seed]

4.4 Why the cut effects are all tiny: the loss-flat tail

Every ΔCE above is $\leq 1.1 \times 10^{-3}$ nats because, beyond lag 2, **P2**’s world is loss-flat. The constrained-belief update assigns each lag- τ term a definite CE value, computable exactly by enumeration.³ The per-lag value ladder is

$$\text{lag 0: } 6.59 \text{ mnat, lag 1: } 1.11 \text{ mnat, lag 2: } 0.11 \text{ mnat, lags } \geq 3: \text{ negative, } \leq 2 \times 10^{-5} \text{ nat.} \quad (14)$$

Removing a far-lag term *helps*: additive evidence over an autocorrelated chain double-counts, so the ζ -kernel’s tail overweights. The whole pattern tail $\tau \geq 3$ is a $\leq 20 \mu\text{nat}$ object. Three consequences, all [measured, exact enumeration]:

- *CE-optimal shrinkage.* The CE-optimal lag kernel is cleanly geometric with rate $0.464 < \zeta = 0.55$ — *faster* decay than the process eigenvalue, the prior sprint’s partial-regression shrinkage, exact in **P2**’s own setting. The trained models track it: on trained rows 4–8 the decay ratio is 0.43–0.45, matching 0.464 and clearly below ζ . **P3**’s own machinery predicts the rate: the MSE-optimal geometric rate for this process (**P3**’s free-profile reduction, matched family) is 0.467, within 1% of the CE optimum — the MSE $\leftrightarrow$ CE gap, the program’s main formal caveat, does not move the optimal spectral rate here.⁴
- *CE-side Toeplitz optimality.* Allowing the kernel to vary freely per destination (55 parameters instead of 10) improves CE by $2.7 \times 10^{-6} < 3 \mu\text{nats}$: a numerically exact CE-side analogue of **P3**’s Toeplitz optimality.
- *The untrained row-9 artifact.* Position 9 predicts token 10, which is never scored, so no gradient ever flows through attention row 9; it is pure noise, an order of magnitude off, with a spurious lag-4 bump. Naive fitted decay rates that pool all rows (including row 9) read 0.68–0.71 and would wrongly reject the spectral ansatz; restricted to trained rows the kernel is clean. Any FRA analysis that attributes meaning to flat-tail wiggles or gradient-free rows is reading noise.

This is the boundary theorem borne out: on separable Mess3 data the optimal pattern carries almost no content-causal mass, so FRA-QK cut effects read gauge plus loss-flat seed noise — the gauge-invariant content of an FRA-QK cut here is indistinguishable from the floor the loss never sees.

The shrinkage weakens with the evidence — a falsifiable corollary. Repeating the ladder at **P2**’s other grid point $\alpha = 0.2$ (weaker emissions; `out/kernel_ce.A20.json`) turns the shrinkage claim into a dependence. The recoverable lag-information falls $\sim 14\times$ (whole ladder ≈ 0.66 mnats, top rung 0.47 mnats, and *no negative rungs* — the tail no longer over-counts), and the CE-optimal geometric rate rises to $0.529 \approx \zeta = 0.55$: the shrinkage nearly *vanishes* at weak evidence, because the double-counting that drives it is second-order in the evidence strength. The ζ -kernel now sits within $1 \mu\text{nat}$ of the family optimum, while the trained kernel wanders further (fitted rate 0.72) exactly as the weaker pinning predicts; the qualitative **P2** picture persists (the attn-out probe still prefers r_1 over η , 0.795/0.769). [measured, second (x, α)]

³`out/kernel_ce.A.json, kernel_ce.py`; clamp-renorm belief readout. ζ -kernel CE 1.0895620, Bayes 1.0892030.

⁴`out/A/close_gaps.json, mse_optimal_rate_scan.unnormalized.`

5 FRA-OV in P2's world

FRA-OV is the *positive control*: unlike FRA-QK, OV cuts causally remove concept content, exactly as the closed forms predict.

5.1 Attribution closed forms

Proposition 5 (attribution closed form; imported [prior-A]/[prior-B]). The per-(token, lag) transported content $\mathbf{A}_{d,s}\mathbf{R}\mathbf{v}_s$ (all lag structure in $\mathbf{A}$, all content in $\mathbf{g}$) is

$$\mathbf{A}_{d,s}\mathbf{R}\mathbf{v}_s = \zeta^{d-s}\mathbf{g}(z_s) \quad (\text{ideal, [exact given ansatz]}), \quad \frac{\zeta^{d-s}}{1-\zeta^d}\mathbf{g}(z_s) \quad (\text{warp-realizable, [exact]}). \quad (15)$$

This is **P1**'s *effective subspace attention* in the SAE basis whose token latents align with $\mathbf{g}(z)$, and by Lemma 2 it is exactly **P3**'s nonstationary regression image. For Config B, per head on its parity class $\Delta_{s \rightarrow d}^{(h)} = \pm \mathbb{K}[\text{parity}] |\zeta|^{d-s} \mathbf{g}(z_s)$ ((9)), and the head-sum is the invariant $\zeta^{d-s} \mathbf{g}(z_s)$.

5.2 Intervention calculus

Severing z^* 's OV channel across sources at gain c edits $\mathbf{c}_d$ only (the softmax is *not* re-run: the edit is linear/clean-pattern, so the intervened activation is exactly affine in c); the skip survives. With the $k=0$ split (7), the modified constrained belief is

$$\mathbf{r}'_1(d; c, z^*) = \mathbf{r}_1(d) - c \left[\underbrace{\sum_{s < d: z_s = z^*} \zeta^{d-s}}_{\text{lags} \geq 1} + \underbrace{a_0(d) \mathbb{K}[z_d = z^*]}_{\text{lag-0 diagonal share}} \right] \mathbf{g}(z^*) \quad (16)$$

— the skip share $(1 - a_0) \mathbb{K}[z_d = z^*] \mathbf{g}(z^*)$ survives, and $a_0(d)$ is the pure-gauge diagonal share (7). [exact given ansatz; verified 2.2×10^{-16} at $c = 1$, 4.4×10^{-16} general c .] A *lag-window* edit restricts the τ -sum to $\tau \in W \subseteq \{0, \dots, L-1\}$ (lag-0 again carrying only its diagonal share): $W = \{1\}$ tests the nearest neighbour, $W = \{2, 3, \dots\}$ the long tail. Both feed the same “edit $\mathbf{c}_d$, keep the rest of the network” pipeline. [exact given ansatz]

5.3 The ex-ante tracking law, verified to four significant figures

Proposition 6 (tracking null and path share; imported [prior-A]/[prior-B]). [tracking law exact; ρ from clean statistics] The projection of the edited belief onto the concept direction $\mathbf{g}(z^*)$ is exactly affine in c : $\psi(c) = \psi_{\text{tot}} - c\psi_S$, $\rho = \psi_S/\psi_{\text{tot}}$, so the concept-nulling gain is $c_{\text{track}}^* = 1/\rho$. The $k=0$ gauge sets ρ : $c^*(a_0) = 1/(\rho_{\text{off}} + a_0 K_0)$ is a property of the run, not the task.

The central positive result is that ρ can be predicted *ex ante* from clean-model statistics alone — no edited forward passes. The ex-ante slope is the joint statistic

$$\rho_{\text{eff}} = \frac{J}{E_z^2 \text{slope}_0}, \quad (17)$$

with J the mean-pattern $\times$ per-cell transported-content $\times$ evidence joint, E_z^2 the evidence normalization, and slope_0 the *clean* per-token tracking slope. The unit bookkeeping is the crux of an honest reading: the intervention reports tracking in *ratio* units (`ov.edits.py` measures `tracking_ratio = slopes/slope_0`), so a raw removal joint must be divided by the clean slope_0 to be comparable. The raw-unit intermediate J/E_z^2 *under-predicts* the ratio-unit measured slope; applying the clean slope_0 conversion recovers it on all five channels.⁵

⁵`out/A/refine_rho.json` (regenerated): `rho_eff_predicted` holds the ratio-unit prediction $J/(E_z^2 \text{slope}_0)$, the raw-unit intermediate J/E_z^2 is stored as `rho_absolute_units_J_over_E2`, and `rho_eff_measured` is the measured ratio-unit slope; $\text{slope}_0 = \text{clean_tracking_slopes}$ from `out/A/ov.edits.json` is computed from clean forward passes only. The null crossing $c^* = 1/\rho_{\text{eff}}$ (oracle z_0 : 0.856) is predicted before any edit.

channel	oracle z_0	oracle z_1	oracle z_2	SAE z_0	SAE z_2
J/E_z^2 (raw joint, <code>rho_absolute_units</code>)	0.877	0.914	0.811	1.379	1.153
$\rho_{\text{eff}} = J/(E_z^2 \text{slope}_0)$ (<code>rho_eff_predicted</code>)	1.168	1.079	2.193	1.838	3.118
measured slope (<code>rho_eff_measured</code>)	1.168	1.079	2.193	1.838	3.118

Because the pipeline is exactly linear in c (frozen pattern, linear readout, pre-MLP; $R^2 = 1.000000$), the ex-ante J/E_z^2 equals the measured raw-removal slope up to the enumeration — the near-exactness is linearity doing its job, a consistency check on the removal mechanism rather than a lucky fit. The one input not predicted from first principles is slope_0 itself: it is applied on both the predicted and measured sides, and $\text{slope}_0 = \{0.750, 0.847, 0.370\}$ comes out *identical* for a token’s oracle and SAE channels — a genuine per-token clean-model quantity (z_2 ’s low 0.370 is not a failure but exactly what makes its ρ large). The null $c^* = 1/\rho_{\text{eff}}$ (oracle z_0 : 0.856) is predicted before measuring; the oracle z_0 tracking ratio moves $+0.71, +0.42, +0.12, -0.17, -0.75$ at $c = 0.25, \dots, 1.5$, removing and *reversing* the token’s own concept tracking. [law + linearity exact ($R^2=1$); ex-ante ρ matches to 4 s.f. once the raw joint is divided by the clean per-token slope_0]

Why $\rho > 1$. The attention channel over-carries relative to ζ -kernel evidence units because the softmax-realizable head output is the constrained update times the normalization warp $\lambda_d = 1/(1 - \zeta^d) \geq 1$ (§3.3) composed with the trained model’s CE-shrinkage (§4.4). The measured $\hat{\lambda}(d)$ declines like the theory warp but sits *below* 1 for $d \geq 2$ (warp $\times$ shrinkage superposed); we report it as the product and do not decompose the two factors quantitatively. The lag-0 term is split diagonal/skip $\approx 85/15$ (tokens 0, 1; $\approx 69/31$ for token 2) — the $k=0$ gauge, measured, skip share position-independent to 7 decimals (0.17948); an OV cut removes only the diagonal share. [explained qualitatively (warp $\times$ shrinkage), not decomposed]

5.4 Quadratic CE with no null; forced collateral

Quadratic CE, no behavioural null. The exact severing CE is $\text{CE}(c) = \text{CE}_0 + c\ell_1 + \frac{c^2}{2}\ell_2$; for $x = 0.15$ the constrained predictor is nearly Bayes-optimal (gap 7×10^{-5} nats), so $\ell_1 \approx 0$ and the CE vertex sits near 0 while the tracking null sits at $1/\rho$ — two distinct predictions. The one-layer model has no redundant second path, so “use” $\equiv$ “presence”: there is no gain-tuned behavioural null in **P2**’s world (the removal factor $\text{RF}(c) \approx (c\rho)^2$ under concept purity; a single-token channel is impure, so c^* must be read from the tracking null, not from $\text{RF}(1)$). [measured]

Geometrically forced collateral. The three displacement directions $\mathbf{g}(z)$ span a 2-plane and sum to zero, so no edit can reduce one token’s coupling without increasing the others’: cutting z_0 boosts z_1, z_2 tracking above 1 (token-1 tracking rises to $\sim 1.5\times$). “Selective concept removal” is impossible here for dimension-counting reasons, before any question of method. [exact given ansatz; measured]

Corollary 1 (geometry collapse). [exact given ansatz; measured] The intermediate fractal $\{\mathbf{r}_1(d) : \text{prefixes}\}$ collapses along $\mathbf{g}(z^*)$ under the z^* cut: the belief-cloud variance along $\widehat{\mathbf{g}(z^*)}$ drops as $(1 - c\rho)^2$ up to a single-feature impurity floor. On the idealized model the plane covariance eigenvalues go isotropic $[0.0731, 0.0731] \rightarrow [0.0098, 0.0731]$ ($7.5\times$, 87% removed) at $c = 1$; on the trained checkpoint (oracle g -channel, exact 3^{10}) clean $[0.097, 0.100] \rightarrow$ cut- z_0 $[0.0158, 0.100]$ (16%), $c = 2$ $[0.0050, \cdot]$ (5%), cut- z_2 $[0.0214, \cdot]$ (21%): one plane eigenvalue collapses, the orthogonal one is untouched. The residual $\sim 16\%$ is the impurity — the $\mathbf{g}$ ’s are non-orthogonal ($\sum_z \mathbf{g} = 0$), so other tokens’ displacements retain a $\widehat{\mathbf{g}(z^*)}$ -component the z^* -only cut cannot remove; full plane collapse needs ≥ 2 token-feature cuts.

Lag-window ladder. Exact-enumeration $\Delta\text{CE}(c = 1)$ for lag-window cuts: lag-0 8.09×10^{-3} , lag-1 3.27×10^{-3} , lag-2 5.40×10^{-4} , lags 3–9 2.74×10^{-4} ; ratios 0.41, 0.17 bracket $\zeta^2 = 0.303$ (each lag carries ζ^τ evidence, CE cost quadratic near the optimum), and the tail is nearly inert. But the $\tau \geq 3$ cost *exceeds* the kernel-family prediction (≈ 0) by $16\times$ because the cut removes the window’s *entire* content (including normalization-pedestal components the MLP relies on), not just its $\mathbf{g}$ -content: full-content window cuts carry built-in collateral. The $\mathbf{g}$ -content-only window cut (now run) lands in the ladder-predicted rank order over 1.5 decades — ΔCE

$1.69 \times 10^{-3} / 3.14 \times 10^{-4} / 6.6 \times 10^{-5}$ for lag-1/lag-2/lags-3-9, vs the ladder’s $1.11 \times 10^{-3} / 1.13 \times 10^{-4} / -4.2 \times 10^{-5}$ — while the full-content cuts above are $\sim 4 \times$ larger: the excess is exactly the non- $\mathbf{g}$ content.⁶ [measured; collateral confirmed]

6 The dictionary theorem-of-practice

The closed forms of §5 presume the analyst can name the token channel $\mathbf{g}(z)$. The sparse dictionary that provides the names is itself the binding constraint, in the cleanest habitat FRA will ever see.⁷

A minimal contrast code; a missing channel. On resid_pre a TopK SAE (12 latents, $K = 3$, FVU 2×10^{-6}) reliably learns 9 *positional* latents and *two* token latents for *three* tokens (all three SAE seeds). **Token 1 is encoded by absence** ($+ \mathbf{b}_{\text{dec}}$), so “sever token 1’s channel” does not exist as a latent-set operation — a concrete instance of the single-feature rank-deficiency of §5.4: the concept is 2-dimensional, the TopK code is minimal. The oracle $\mathbf{g}(1)$ channel exists for all three tokens and its cut works ($\Delta\text{CE } 3.46 \times 10^{-3}$), so the obstruction is the code’s minimality, not the concept’s causal reachability. [measured]

Channels carry contrasts, not theory directions. The latents that exist decode as contrasts: the summed token-0 feature is $1.01 \mathbf{e}(0) - 0.93 \mathbf{e}(1)$, and its transported belief content matches the code-predicted mixture $\chi \cdot \mathbf{g}$ at $\cos 0.9992/0.9953$ (vs $0.90/0.82$ against pure $\mathbf{g}(z)$), where χ is the mixing matrix (support-restricted estimate: rows $u_0 \approx (1.01, -0.93, 0.00)$, $u_2 \approx (0.01, -0.93, 1.00)$ in the $\mathbf{e}$ -basis, token 1 the default). For reference the *oracle* centered token content $\mathbf{e}_c(z)$ is parallel to $\mathbf{g}(z)$ at $\cos 1.0000/0.9966/0.9997$. [measured]

Purity is necessary, not sufficient. Set-level token purity is perfect ($R^2 \approx 1.0$); purity — the prior sprint’s criterion — is not the obstruction. **Completeness and centering of the code are what FRA edits inherit:** all the intervention numbers of §5 remain exactly predictable, but only through χ , which the analyst must extract from the SAE first (this is why the SAE-channel ρ_{eff} in the §5.3 table is larger than the oracle’s — the contrast content plus pedestal). The two-head model ($\zeta < 0$) is worse: its head-summed channel geometry scrambles (per-head OV’s anti-parallel, so only per-head or pattern-weighted objects are meaningful), and even the *number* of token latents varies across SAE seeds ($4/3/2$). The resid_mid SAE does not hide the computation: its error carries *less* belief-plane content (3.3%) than its overall size (5.9%). [measured]

7 The two-head hinge ($\zeta < 0$)

7.1 What the loss pins

For $\zeta = -0.5$ softmax nonnegativity forces two heads with anti-parallel OV’s; only the head-sum is loss-pinned,

$$\sum_{h=1,2} \mathbf{A}_{d,s}^{(h)} \mathbf{R} \mathbf{v}_s^{(h)} = \zeta^{d-s} \mathbf{g}(z_s) = \text{pinned}(\tau) \mathbf{g}(z_s), \quad \text{pinned}(\tau) := \zeta^\tau. \quad (18)$$

The invariant is this head-summed FRA-OV attribution; per-head attributions are gauge.

Proposition 7 (per-head gauge family and its softmax obstruction). [exact in M1; approximate under softmax] Parametrize head h by a nonnegative lag profile $a_h(\tau)$ and signed OV gain γ_h ($\gamma_1 > 0 > \gamma_2$) with $\gamma_1 a_1(\tau) + \gamma_2 a_2(\tau) = \zeta^\tau$; the freedom is a *common additive profile* $\delta(\tau) \geq 0$ added to both heads (plus a magnitude split), with the canonical (**P2**) split $a_1^{\text{can}}(\tau) = |\zeta|^\tau \mathbb{K}[\tau \text{ even}]$, $a_2^{\text{can}}(\tau) = |\zeta|^\tau \mathbb{K}[\tau \text{ odd}]$ the minimal member ($\delta \equiv 0$). The family has dimension L (one half-line $\delta(\tau)$ per lag) plus one magnitude split, and is loss-preserving *exactly* for free row-stochastic patterns (**P3**’s M1). Under softmax with per-head row normalization the transformation

⁶out/A/close_gaps.json: gcontent_window_dCE.c1 vs ladder_prediction.

⁷out/A/fra_dict.json, out/A/dict_deep.json; fra_dict.py, dict_deep.py. resid_pre is attention’s input, a deterministic function of (token, position) with 30 distinct values.

$\sigma^{(h)}(\tau) \rightarrow \log(e^{\sigma^{(h)}(\tau)} + \delta(\tau))$ (applied identically to both heads, OV blocks untouched) is only approximately loss-preserving: $O(\varepsilon)$ on rows where both heads have support (coefficient ≈ 0.19 per ε , measured), but $O(1)$ at rows where the common profile newly activates a head whose parity leaves it keyless — most sharply at $d = 1$, where a lone new entry renormalizes to 1 and the head-sum changes by a full $\mathbf{g}$ -vector (0.2667, independent of ε). The only *exact* per-head dials are head relabeling (with OV sign, verified 0.0) and the per-head $k=0$ gauge; no exact per-head OV-magnitude split exists for $\zeta < 0$ (that would need the two heads to share a pattern, but a shared nonnegative pattern cannot produce the sign oscillation ζ^τ).

7.2 Measured verification (Config B / Brich)

We use **Brich** (rich init, `init_range` 0.02) as the cleanest B model: token-dependence relstd 0.056, anti-parallel OV cosine $-0.98/-0.98/-1.0$.⁸

Head-sum recovers the oscillation; per-head is single-signed. On Brich, each head’s effective subspace attention is single-signed (OV along $+\mathbf{g}$ or $-\mathbf{g}$), so the sign oscillation cannot live in either head; the head-sum recovers the signed ζ^{d-s} kernel — cosine 0.985 on the exact row-8 per-cell measurement, and 0.994 across training on the lag-1–7 profile estimator (`twohead.json`’s `headsum_cos_zeta_lags1-7`; lag 0 excluded, as the skip/diagonal split depresses it). [confirms (18)]

The invariant locks on; per-head keeps drifting at flat loss. Across training (checkpoints every 500 steps, all three B seeds), the head-sum cosine with ζ^τ (lags 1–7) converges — Brich $-0.169 \rightarrow 0.986$ (mid-training) $\rightarrow 0.994$ (final); B 0.930, B2 0.875 at the end — while the per-head parity purity *keeps rising* at nearly constant CE (head-1 even-lag fraction $0.59 \rightarrow 0.71$ over the final 7k steps). A controlled δ -dial confirms the family is a low-loss direction: per-head $|\Delta\alpha|$ up to 0.23 at $\Delta\text{CE} \leq 1.5 \times 10^{-4}$ (an order below the $\sim 10^{-3}$ loss scale, and *slightly negative* at small δ). [confirms Prop. 7]

Rich/lazy: init decides whether even the invariant converges. With TransformerLens-default init the invariant itself is still climbing at the end of **P2**’s training budget (head-sum cos 0.88–0.93: B2 0.875, B 0.930); rich init (Brich 0.994) locks it early. This is **P1** App.-G rich/lazy reproduced in **P2**’s world: per-head FRA rankings are checkpoint- and init-dependent even when seeds agree on the canonical parity split. [measured]

8 The factored world (Phase 2, P1 App. L)

Phase 2 is **P1**’s factored two-Mess3 product $\mathbf{T}^{(x)} = \mathbf{T}_1^{(z^{(1)})} \otimes \mathbf{T}_2^{(z^{(2)})}$ (9 latent states, 9 tokens, $d_{\text{model}} = 120$, sequence length 11 *with* BOS; `train_p1.py`). The configurations are P1mix2 ($\zeta = (+0.5, -0.5)$, $H = 2$), P1pos2 ($(+0.7, +0.4)$, $H = 2$), P1mix3 ($H = 3$), 100k steps each. The Phase-1 machinery lifts directly, and the gauge story becomes richer in three ways.

8.1 Theory statements

1. *Per-factor constrained update.* Each factor n has its own $\Delta\mathbf{r}_n = \sum_s \zeta_n^{d-s} \mathbf{g}_n(z_s^{(n)})$ in a $\approx$ orthogonal 2-d residual subspace V_{F_n} (**P1**’s vary-one identification). The FRA-OV closed forms (§5) apply *per subspace*: attribution $\mathbf{A}_{d,s} \mathbf{R}_n \mathbf{v}_s = \zeta_n^{d-s} \mathbf{g}_n(z_s)$, projected by **P1**’s *effective subspace attention* $\alpha_n(d, s)$ — exactly “FRA-OV aggregated to the known subspace $\mathbf{g}_n$ ”. The invariant is now the head-sum *per subspace*; per-head-per-factor is gauge. [conjecture, lift of §5]
2. *Ray reuse and the conic head count.* The negative- ζ two-ray requirement (§3.4) becomes **P1**’s conic minimum head count $H_{\min}$: the fewest heads whose nonnegative cone contains the lag-update set $\{(\zeta_1^k, \zeta_2^k)\}$. Rays can be *reused across factors* — e.g. $\zeta_1 = +0.5, \zeta_2 = -0.5$ needs 2 heads, not 3 — so the per-head

⁸`out/twohead.json, p2_checks.json; scratchpad/verify_2head.py`, exact 3¹⁰.

gauge family (Prop. 7) grows: a head can serve two factors, and δ -freedom now includes cross-factor re-splits. The head-summed-per-subspace attribution is still the only loss-pinned object. [conjecture]

3. *The QK gauge survives verbatim.* **P1**'s causal shuffles (token-identity shuffles of QK inputs leave the loss unchanged; lag-breaking shuffles hurt) are exactly the §4 statement that FRA-QK carries *temporal structure only*; the tok×tok block is gauge. The G1/ δ dials transfer; the new subspace structure only adds per-factor pedestals β_n and shares θ_n . [conjecture, lift of §4]

Factor-selective interventions and the orthogonal-collateral contrast. Because the factor subspaces $V_{F_1} \perp V_{F_2}$ are (approximately) orthogonal — unlike the three coplanar $\mathbf{g}(z)$ of a single Mess3 — a cut of factor 1's channel should produce *no* forced collateral on factor 2, in sharp contrast to Mess3's geometrically forced collateral (§5.4). This is the cleanest predicted discriminator between the factored and single-process worlds: selective concept removal is *possible per factor* where it was impossible per token. The BOS token adds a genuine attention sink (unlike **P2**'s sink-at-position-1 workaround, Prop. 3), which should *absorb* the normalization warp and make the per-subspace α_n cleaner than **P2**'s warped head output. [conjecture]

8.2 Measured facts (final models)

All numbers below are from the fully trained models (100k steps; Rao–Blackwellized Bayes gaps $1.7\text{--}2.0 \times 10^{-4}$ nats — all three train essentially to Bayes).⁹

- *Per-factor effective subspace attention matches theory — both eigenvalues from one model.* Distinct-positive ($\zeta = (+0.7, +0.4)$, $H=2$): α_1 lag ratio +0.705, α_2 ratio +0.381 (cos with ζ_n^τ over lags 1–7: 0.9971, 0.9995). Mixed ($(+0.5, -0.5)$, $H=2$): ratios +0.572 / -0.570 (cos 0.9945/0.9930), including the sign oscillation of the negative factor — which appears in *neither* head's own α (per-head α is single-signed; the oscillation lives in the head-sum). Statement 1 of §8.1 is thereby [measured], not conjecture.
- *Head-factor shares separate the regimes.* Distinct-positive *specializes*: share matrix $\begin{bmatrix} 0.95 & 0.28 \\ 0.05 & 0.72 \end{bmatrix}$ (rows heads, columns factors), sharpening from 0.71 at 5k steps. Mixed *collaborates*: $\begin{bmatrix} 0.64 & 0.64 \\ 0.36 & 0.36 \end{bmatrix}$ — identical proportions across factors (head roles are parity-based, not factor-based, when a negative eigenvalue forces the two-head split). The $H=3$ mixed model leaves its third head with a share below 5×10^{-4} of both factors (its softmax pattern still sums to 1, but its OV transports $\sim 10^3 \times$ less factor content than the live heads) — a dead head in the FRA-OV sense: the conic $H_{\min} = 2$ with ray reuse, made visible. Conversely, the $H=1$ mixed model bounds $H_{\min}$ from below: with a single head its RB gap plateaus at 4.9×10^{-4} nats ($2.8 \times$ the $H=2$ gap) and the shared α kernel *truncates at lag 1* (lag-2/lag-1 ratio 8×10^{-4} for both factors) — a single head gives the two factors' transported coefficients a lag-independent ratio γ_2/γ_1 , whereas the target ratio $(-0.5)^\tau/(0.5)^\tau = (-1)^\tau$ alternates in sign with lag, so only one lag can be served exactly; training keeps the dominant lag-1 evidence and zeroes the tail.¹⁰
- *Factor-selective severing with a built-in collateral control.* Severing factor n 's OV-transported update drives factor- n tracking down linearly while factor m 's tracking reads +1.00 at every gain in every model (ex-ante prediction: 1.00 ± 0.05): orthogonal factor subspaces remove the forced collateral of single Mess3 (§5.4). The matching key-side FRA-QK cuts cost -1×10^{-4} to -3.4×10^{-4} nats (slightly *negative*), gain-independent, tracking pinned at 1.00, while the OV cuts reach 1.5×10^{-2} nats and null the tracking: the §4 gauge story in factored form.
- *The $k=0$ gauge sets the severing reach — across configurations.* The mixed model severs $\rho \approx 0.35$ of tracked factor content per unit gain; the distinct-positive model $\rho \approx 0.97$ (tracking 0.03 at $c=1$; null at $c \approx 1.03$). Embedding factorization is *not* the mechanism (interaction share 1.1% in both; hypothesis tested and refuted). The mechanism is the lag-0 delivery route: mixed delivers lag-0 factor content via the *skip* (skip 0.88 / diagonal 0.07), distinct-positive via *diagonal attention* (skip 0.05 / diagonal 1.02), and at

⁹out/{P1mix2,P1pos2,P1mix3}/analyze.json and factor_edits.json; analyze.p1.py, factor_edits.py, sample $N=4000$ with exact per-factor Bayes conditionals (RB estimator). The intervention predictions were logged *ex ante* in RESEARCH_LOG.md (t $\approx$ 2h25) before the battery ran.

¹⁰out/P1mix1/analyze.json: gap.to.bayes 4.93×10^{-4} , ratio.lag2.over.lag1 $\approx 8 \times 10^{-4}$ for both factors.

$|\zeta| = 0.5$ lag 0 carries roughly half the evidence weight. This places the two models at opposite ends of the same $k=0$ gauge orbit — the $\rho(a_0)$ law of §5.3 (“ c^* is a property of the run”) as an *inter-model* fact. Two further measurements show the chosen end is *spectrum-determined*, not SGD luck. First, a second seed per config picks the *same* end (mixed seed 43: skip 0.85/diag 0.08, ρ 0.42/0.32; distinct-positive seed 43: skip 0.06/diag 1.03, ρ 0.97/0.95), with α -ratios, regimes and RB gaps replicating up to a $\sim 20\%$ magnitude spread. Second, across six models spanning both phases, seeds and init scales, a purely positive spectrum takes the *diagonal* end (**P2** config A, $\zeta = +0.55$: diag 0.78/skip 0.18) while *any* negative eigenvalue takes the *skip* end (Brich, $\zeta = -0.5$: skip 0.73/diag 0.27; config B: skip 0.90/diag 0.13), corroborated by Brich’s intervention profile (oracle $\rho \approx 0.3$, its diagonal-window cut $29\times$ cheaper than A’s). The mechanism is now *tested* by a binding no-skip intervention (remove the residual skip around attention at training time, so lag 0 can arrive *only* via diagonal attention — unlike a frozen embedding, an absent path cannot be read around): parity-split heads make the diagonal an even-lag slot whose loading couples lag 0 to the even head’s row normalization (which must also produce the $|\zeta|^{2k}$ ladder), whereas the skip delivers lag 0 outside every softmax normalization, so a sign-oscillating spectrum routes lag 0 to the skip. Forcing the negative model onto the diagonal costs +0.091 mnats (gap 0.442 vs Brich’s 0.351, 26% worse) and visibly strains one head — head 1 carries diagonal mass 0.52 while its even-lag ladder compresses to 0.11 in the same row normalization, the diagonal $\leftrightarrow$ ladder tension in one number pair — whereas the *positive* model pays nothing (0.258 vs 0.259 mnats, though the same edit also removes its direct positional path: a strong control). [tested: binding no-skip intervention, one seed per config] The $\rho(a_0)$ law thereby sharpens from “ c^* is a property of the run” to *the run’s position on the $k=0$ orbit is itself predictable from the data spectrum*.¹¹

- *The dictionary does not factorize where the model most needs it.* A TopK SAE on the mixed model’s `resid_pre` (24 latents, $K=4$, FVU 7.5×10^{-3}) gives factor 1 a complete per-value code (3 latents) but the negative factor 2 a *single* latent across its three values, with 11/24 latents mixed (`sae_p1.json`): the §6 completeness obstruction, selecting against precisely the factor whose representation is split across anti-parallel heads.
- *Belief-plane recovery and the BOS sink.* Factored beliefs explain 0.71–0.84 of centered residual variance; factor subspaces are near-orthogonal (largest principal-angle cosine 0.097). The head-summed BOS mass decays with log-slope $-0.41/-0.43$ across rows for the $H=2$ models but is nearly flat (-0.04) for $H=3$: consistent with an active sink but *not* a clean quantitative confirmation of the warp-absorption prediction; we report it as an observation.

All Phase-2 numbers are [measured, final models].

8.3 Outlook: the two-eigenvalue optimal profile (P3 lift)

The factored process places **P3**’s optimal-attention analysis in its first genuinely multi-eigenvalue setting. The product transition $\mathbf{T} = \mathbf{T}_1 \otimes \mathbf{T}_2$ has nonstationary spectrum $\{\zeta_1, \zeta_2, \zeta_1\zeta_2\}$, and the token displacement carries all three: $\mathbf{g}(j) = \mathbf{g}_1(j_1) \otimes \boldsymbol{\pi}_2 + \boldsymbol{\pi}_1 \otimes \mathbf{g}_2(j_2) + \mathbf{g}_1(j_1) \otimes \mathbf{g}_2(j_2)$ [exact]. The cross-eigenvalue $\zeta_1\zeta_2$ therefore *enters* the skip-bigram statistics through the interaction term $\mathbf{g}_1 \otimes \mathbf{g}_2$ (subdominant, $\|\mathbf{g}_1 \otimes \mathbf{g}_2\| \sim \|\mathbf{g}_2\|/\|\boldsymbol{\pi}_2\|$ times the marginal parts): factor independence factorizes the beliefs but not the joint displacement, since one token constrains both factors. **P3**’s force-balance/generating-function route then predicts an optimal profile that is a sum of *three* geometrics, the in-disc roots of a degree-6 reciprocal polynomial (the generalization of **P3**’s single-eigenvalue palindromic quadratic; the coupled roots are not in closed form and we compute them numerically). Per factor the MSE optimum is the **P3**-shrunk rate ($\eta^* = 0.60, 0.34$ for $\zeta = 0.7, 0.4$, ratio 0.85, reproducing **P2**’s 0.467 at $\zeta = 0.55$), *but* the shrinkage is loss-flat: using the raw ζ costs 2.2%/0.56% of the recoverable lag-information ($6.9 \times 10^{-5}/1.3 \times 10^{-6}$ MSE). This explains why the trained per-subspace α_n decays at $\approx \zeta_n$ (measured 0.705/0.381, §8.2) rather than at η^* : the decay rate is the factored-world instance of the loss-flat regime of §4.4, a near-gauge quantity below the CE gradient floor. The $\zeta_1\zeta_2$ geometric contributes a real but $\sim 2 \times 10^{-5}$ improvement — present in the theory, invisible to training.¹² [exact spectrum; numerical rates; analytical frontier = coupled roots]

¹¹Route splits out/{A,Brich,B}/skip_diag.p1.json and out/{P1mix2s43,P1pos2s43}/skip_diag.json; Brich intervention out/Brich/ov_edits.json; no-skip test out/noskip.json.

¹²Full derivation and numerics: `derivations.p2_fra.md` §9 and `scratchpad/ext_b.py`.

8.4 Outlook: the mixture ($\lambda \rightarrow 1$) limit

The one regime this sprint deliberately left out is the prior sprint’s mixture world — a concept carried by a non-decaying ($\lambda \rightarrow 1$) mode. Every object measured here extrapolates to that limit. As $\zeta \rightarrow 1$ the per-lag value ladder of §4.4 flattens (rung ratios scale as $\zeta^{2\tau}$), so the loss-flat tail that begins at lag 3 for $\zeta = 0.55$ recedes to infinity: the *entire* profile becomes loss-pinned, toward the flat optimum of the imported flattening law ([prior-A]/[prior-B]: $1/(1-\eta) \propto (1-\lambda)^{-1/2}$). The $k=0$ route question dissolves — lag 0 becomes one more equal rung of a counting kernel — and the FRA handles inherit their known aggregation-regime roles: the pattern is maximally pinned yet carries no concept (the normalization pin $\tilde{\alpha}(1) = 1$, [prior-B]), so FRA-QK stays gauge for the concept, while FRA-OV aggregation shares — and hence c^* — are set by the multi-path structure that the one-layer worlds studied here lack. Measuring the ladder, the shrinkage, and the route law along the path $\zeta \rightarrow 1$ (here sampled at 0.4, 0.5, 0.55, 0.7) is the natural next sprint. [outlook; no new measurements]

9 Honesty ledger

We reproduce `summary.md`’s ledger faithfully; every imported result is attributed and every measured number is traceable to a named JSON artifact.

- The trained pattern’s content leak is large ($\varepsilon_{C1} \approx 1.5\times$ the profile slope): the gauge theorem’s premise holds only *on average*, and the $\sim 10^{-5}$ -nat residue at the dial bottom (§4.3) is a *measured* floor, not a proven zero.
- $\rho_{\text{eff}} > 1$ is *explained* by warp $\times$ shrinkage but not decomposed into the two factors quantitatively; the token-2 clean tracking slope (0.37 vs 0.75/0.85) is anomalously low and unexplained — it inflates that channel’s ρ and c^* without affecting the prediction agreement (the slope₀ division is applied on both the predicted and measured sides).
- The kernel-family prediction for full-content window cuts under-predicts by $16\times$ (collateral of non-**g** content); the **g**-content-only window cut, now run, tracks the ladder in rank order while the full-content cut is $\sim 4\times$ larger — confirming the excess is non-**g** content (§5.4).
- CE-optimal-kernel statements use the clipped belief readout, not the trained MLP; the trained model’s own readout differs (it beats that family’s optimum readout on Config A).
- Two-head α -ratio estimators are noisy near sign changes; parity-share and cosine metrics are the reliable ones.
- The Phase-2 $k=0$ /route $\leftrightarrow$ spectrum result replicates across two seeds per config (same orbit end), so its *direction* is not seed-contingent; magnitudes still carry a $\sim 20\%$ seed spread, and the Phase-1 route anchors (A, B, Brich) are single-seed for this specific measurement.
- The no-skip mechanism test is one seed per config; an earlier probe that only *froze* the token embeddings was inconclusive (the model reads lag-0 content around a path that is still present), which is why the binding version removes the skip outright.
- All Phase-1 claims are single-architecture (**P2** App. B) and Mess3-specific — though the gauge dial is now checked on a second seed (§4.3) and the shrinkage on **P2**’s second α grid point (§4.4); $L = 10$ truncates the warp/ladder at $\zeta^{10} \approx 0.25\%$.
- All results are [exact given the **P2** spectral ansatz]; the ansatz itself (pattern $\propto \zeta^{d-s}$, values/embeddings $\propto \mathbf{g}(z)$) is **P2**’s empirical fit, not derived from CE (Conjecture Z1). The MLP (nonlinear warp to full Bayes) is entirely outside this analysis.

Summary of the dichotomy. On separable constrained-belief data, **FRA-QK attribution measures gauge** — dialable over a $140\times$ range at bit-identical loss, its invariant content at the noise floor — while **FRA-OV attribution is the computation**: every tracking number is predicted *ex ante* to four significant figures, the CE follows the quadratic path calculus, and the only obstructions are the geometrically forced collateral of a single Mess3 and the minimality of the learned dictionary. The two-head hinge and the factored world show where the clean picture bends: per-head decompositions are a gauge family that softmax does not freely realize, and only the head-summed, subspace-projected attribution is a trustworthy handle.